Kuboidning qirrasini topish

Qutining hajmi va ikkita o'lchami ma'lum bo'lsa, uchinchi o'lchamini chizg'ichsiz ham topsa bo'ladi. Buning uchun hajm formulasini «teskari» ishlatamiz.

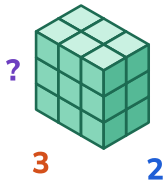
1-QISM

Mavzu

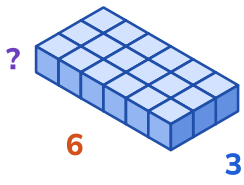
ESLAB OLAYLIK

- Kuboid hajmi = bo'yi × eni × balandligi. Masalan, $5 \cdot 3 \cdot 4 = 60$ cm³. Kub hajmi = qirrasi × qirrasi × qirrasi.
- Noma'lum ko'paytuvchi: $6 \cdot \square = 42$ bo'lsa, $\square = 42 : 6 = 7$.
- $1\ \text{l} = 1000$ cm³. Idishdagi suv hajmi = idish bo'yi × eni × suv sathining balandligi.

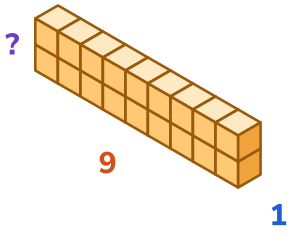
18 ta birlik kubdan uch xil kuboid yasaymiz. Har birida bo'yi va eni ma'lum. Balandligi nechaga teng? Bitta qatlamda bo'yi × eni ta kubcha bor. 18 ni shu songa bo'lsak, qatlamlar soni — balandlik chiqadi.



A



B



C

Kuboid	Bo'yi	Eni	Bir qatlamda	Balandligi	Hajmi
A	3	2	$3 \cdot 2 = 6$	$18 : 6 = 3$	18
B	6	3	$6 \cdot 3 = 18$	$18 : 18 = 1$	18
C	9	1	$9 \cdot 1 = 9$	$18 : 9 = 2$	18

QOIDA · HAJMI VA IKKITA QIRRASIGA KO'RA UCHINCHI QIRRANI TOPISH

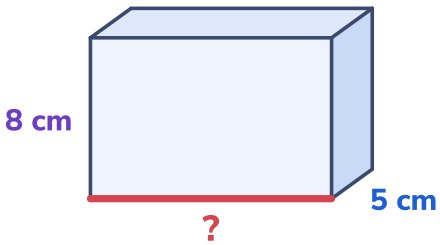
$$\text{Kuboidning balandligi} = \frac{\text{hajmi}}{\text{bo'yi} \times \text{eni}}$$

$$\text{Kuboidning bo'yi} = \frac{\text{hajmi}}{\text{eni} \times \text{balandligi}}$$

$$\text{Kuboidning eni} = \frac{\text{hajmi}}{\text{bo'yi} \times \text{balandligi}}$$

1-MISOL · BO'YINI TOPISH

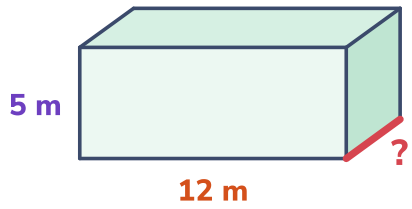
Kuboidning hajmi 480 cm³, eni 5 cm, balandligi 8 cm. Bo'yini toping.



1. Bo'yi $= \frac{480}{5 \cdot 8} = \frac{480}{40} = 12$ cm.
2. ✓ Tekshirish: $12 \cdot 5 \cdot 8 = 480$ cm³.

2-MISOL · ENINI TOPISH

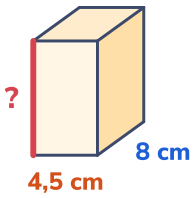
Kuboidning hajmi 360 m³, bo'yi 12 m, balandligi 5 m. Enini toping.



1. Eni $= \frac{360}{12 \cdot 5} = \frac{360}{60} = 6$ m.
2. ✓ Tekshirish: $12 \cdot 6 \cdot 5 = 360$ m³.

3-MISOL · QIRRALARDAN BIRI O'NLI KASR

Kuboidning hajmi 288 cm³, bo'yi 4,5 cm, eni 8 cm. Balandligini toping.



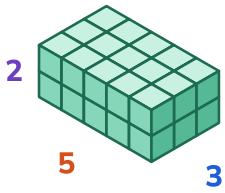
1. Asosdagi ko'paytma: $4{,}5 \cdot 8 = 36$.
2. Balandligi $= \frac{288}{36} = 8$ cm.
3. ✓ Tekshirish: $4{,}5 \cdot 8 \cdot 8 = 288$ cm³.

4-MISOL · JAVOB O'NLI KASR

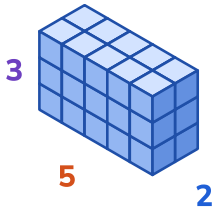
Pechenye qutisiga 1200 cm³ pechenye sig'adi. Qutining tubi 16 cm ga 10 cm. Quti qanchalik baland?

1. Balandligi $= \frac{1200}{16 \cdot 10} = \frac{1200}{160}$.
2. $1200 : 160 = 7$ (qoldiq 80), $80 : 160 = 0{,}5$ (yarim). Javob: $7{,}5$ cm.
3. ✓ Tekshirish: $16 \cdot 10 \cdot 7{,}5 = 1200$ cm³.

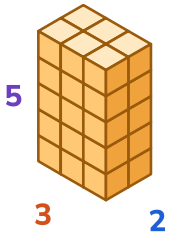
Endi bitta kuboidni uch xil qo'yib ko'ramiz. U 30 ta birlik kubdan yasalgan.



1-holat



2-holat



3-holat

Holat	Asosi	Asosining yuzi	Balandligi	Hajmi
1	5 × 3	15	2	$15 \cdot 2 = 30$
2	5 × 2	10	3	$10 \cdot 3 = 30$
3	3 × 2	6	5	$6 \cdot 5 = 30$

Bir xil kuboid 3 xil asos yuzi va balandlikka ega bo'lishi mumkin. Hajmi esa o'zgarmaydi.

QOIDA · HAJMI VA ASOSINING YUZIGA KO'RA BALANDLIKNI TOPISH

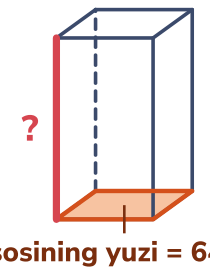
Kuboidning pastki yog'i uning asosidir.

Kuboid asosining yuzi = bo'yi × eni

$$\textbf{Kuboid balandligi} = \frac{\text{hajmi}}{\text{asosi yuzi}}$$

5-MISOL · ASOSINING YUZI BERILGAN

Kuboid asosining yuzi 64 cm², hajmi 960 cm³. Kuboidning balandligini toping.



1 Balandligi $= \frac{960}{64} = 15$ cm.

✓ Tekshirish: $64 \cdot 15 = 960$ cm³.

USUL · BIR YOG'INING YUZI BERILGAN

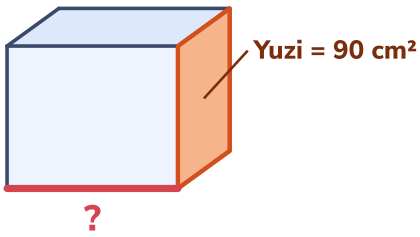
$$\text{Bo'yi} = \frac{\text{hajmi}}{\text{bo'yalgan yog'ining yuzi}} \quad \text{yoki} \quad \text{Eni} = \frac{\text{hajmi}}{\text{bo'yalgan yog'ining yuzi}}$$



Bo'yalgan yoqning yuzi ikki qirraning ko'paytmasiga teng. Shu ikki qirrani bilmasak ham, hajmni yoq yuziga bo'lsak, uchinchi qirra chiqadi.

6-MISOL · YON YOG'I BO'YALGAN

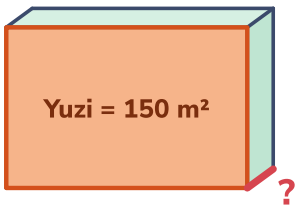
Kuboidning hajmi 1080 cm³. Bo'yalgan yog'ining yuzi 90 cm². Noma'lum qirrani toping.



1. Bo'yalgan yoq eni va balandligidan iborat. Noma'lum qirra — bo'yi.
2. $\text{Bo'yi} = \frac{1080}{90} = 12$ cm.

7-MISOL · OLD YOG'I BO'YALGAN

Kuboidning hajmi 600 m³. Bo'yalgan yog'ining yuzi 150 m². Noma'lum qirrani toping.



1. Bo'yalgan yoq bo'yi va balandligidan iborat. Noma'lum qirra — eni.
2. $\text{Eni} = \frac{600}{150} = 4$ m.

2-QISM

Misollar ishlash

1-MISOL · JADVALNI TO'LDIRISH

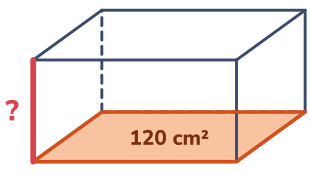
Jadvaldagi bo'sh kataklarni to'ldiring.

Kuboid	Bo'yi (cm)	Eni (cm)	Balandligi (cm)	Hajmi (cm³)
P	7	3	2	42
Q	8	4	5	160
R	15	12	3	540
S	2,5	4	6	60

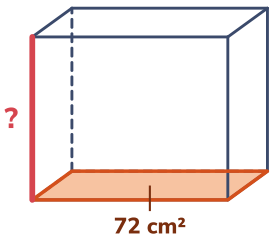
1. Hajm bo'sh bo'lsa — uchala qirrani ko'paytiramiz. P: $7 \cdot 3 \cdot 2 = 42$.
2. Qirra bo'sh bo'lsa — hajmni qolgan ikki qirraning ko'paytmasiga bo'lamiz. Q: $\frac{160}{8 \cdot 5} = \frac{160}{40} = 4$. R: $\frac{540}{12 \cdot 3} = \frac{540}{36} = 15$.
3. S: $2{,}5 \cdot 4 = 10$, $\frac{60}{10} = 6$.

2-MISOL · BITTA QUTI, UCH XIL QO'YILGAN

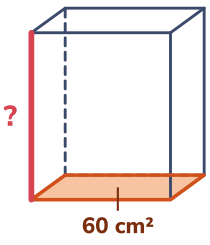
Hajmi 720 cm³ bo'lgan quti stolga uch xil qo'yildi. Asosining yuzi birinchi holatda 120 cm², ikkinchisida 72 cm², uchinchisida 60 cm². Har holatda qutining balandligi qancha?



1-holat



2-holat



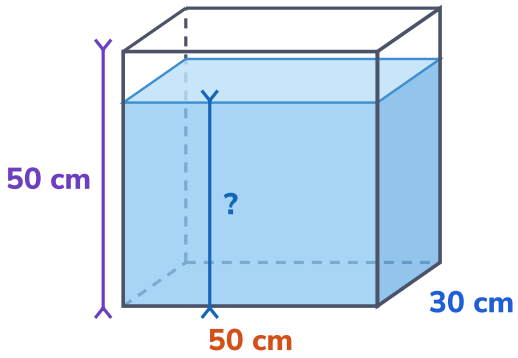
3-holat

1. Hajm har holatda bir xil — 720 cm³. O'zgaradigani faqat asos va balandlik.
2. $\frac{720}{120} = 6$ cm, $\frac{720}{72} = 10$ cm, $\frac{720}{60} = 12$ cm.

Uchta balandlik — qutining uchta qirrasi: 12 cm, 10 cm va 6 cm. Tekshirish: $12 \cdot 10 \cdot 6 = 720$ cm³. Asos kattaroq bo'lsa, balandlik kichikroq chiqadi.

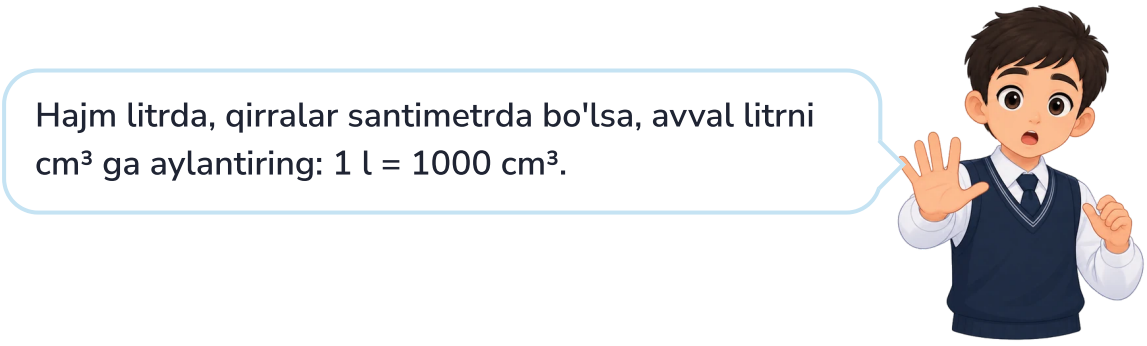
3-MISOL · HAJM LITRDA BERILGAN

Akvariumning bo'yi 50 cm, eni 30 cm, balandligi 50 cm. Unga 60 l suv quyildi. Suv sathi qanday balandlikda?



1. Litrni cm³ ga aylantiramiz: $60\ \text{l} = 60 \cdot 1000 = 60\,000$ cm³.
2. Suv egallagan qism ham kuboid. Uning asosi — akvarium tubi: $50 \cdot 30 = 1500$ cm².
3. Suv sathi $= \dfrac{60000}{1500} = 40$ cm.

Agar savol «akvarium to'lishi uchun yana necha litr suv kerak?» bo'lsa: bo'sh qism balandligi $50 - 40 = 10$ cm, hajmi $1500 \cdot 10 = 15\,000$ cm³ $= 15$ l.



DIQQAT!

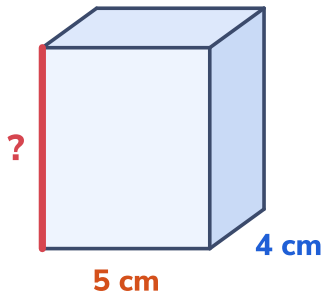
1. Avval ikki qirrani ko'paytiring, keyin hajmni shu songa bo'ling: ✗ $480 : 5 \cdot 8 = 768$ ✓ $480 : (5 \cdot 8) = 480 : 40 = 12$
2. Hajmni qirralar yig'indisiga bo'lmang: ✗ $480 : (5 + 8)$ ✓ $480 : (5 \cdot 8)$
3. Qirra — cm da, yuz — cm² da, hajm — cm³ da yoziladi: ✗ balandligi 15 cm³ ✓ **balandligi 15 cm**
4. Hajm litrda berilsa, avval cm³ ga aylantiring: ✗ $60 : 1500$ ✓ $60\,000 : 1500 = 40$ **cm**

3-QISM

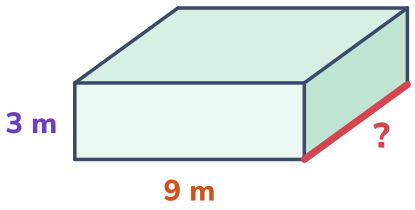
Mustahkamlash

1. Hajmini toping.
   a) Qirrasi 5 m bo'lgan kub. b) Bo'yi 9 cm, eni 4 cm, balandligi 3 cm bo'lgan kuboid.
   c) Qirrasi 7 cm bo'lgan kub. d) Bo'yi 12 m, eni 5 m, balandligi 2 m bo'lgan kuboid.

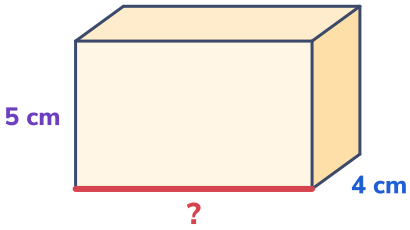
2. Rasmga qarab kuboidning noma'lum qirrasini toping.



a) Hajmi 120 cm^3



b) Hajmi 270 m^3



c) Hajmi 160 cm^3

3. Jadvalni daftaringizga chizib to'ldiring.

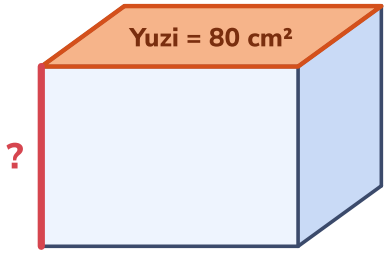
Kuboid	Bo'yi (cm)	Eni (cm)	Balandligi (cm)	Hajmi (cm^3)
A	9	5	2	?
B	12	5	?	300
C	15	?	6	540
D	?	8	2,5	220
E	20	20	?	2000

4. 40 ta birlik kubdan uch xil kuboid yasaldi. Noma'lum o'lchamni toping.
   a) bo'yi 5, eni 4, balandligi ?; b) bo'yi 10, eni ?, balandligi 1; c) bo'yi ?, eni 2, balandligi 2.

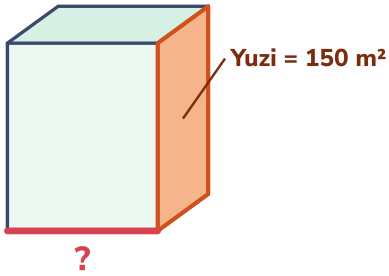
5. Kuboid shaklidagi sovun bo'lagining bo'yi 9 cm, eni 6 cm, hajmi 162 cm^3. Sovunning qalinligi (balandligi) qancha?

6. Kuboid shaklidagi muz bo'lagi asosining yuzi 36 cm^2, hajmi 324 cm^3. Muz bo'lagining balandligini toping.

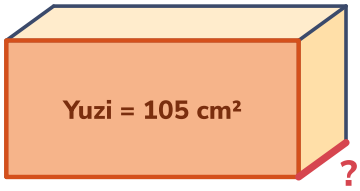
7. Kuboidning bo'yalgan yog'i yuziga ko'ra noma'lum qirrasini toping.



a) Hajmi 560 cm³

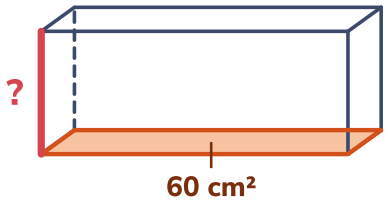


b) Hajmi 1800 m³

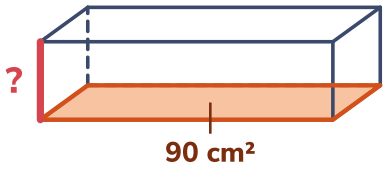


c) Hajmi 630 cm³

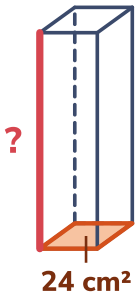
8. Hajmi 360 cm³ bo'lgan o'yinchoq yog'och g'ishtcha uch xil qo'yildi. Har holatda asosining yuzi rasmda yozilgan. Har holatda balandlikni toping. G'ishtchaning qirralari qancha?



1-holat



2-holat



3-holat

9. Basseynga 450 m³ suv quyildi. Basseynning bo'yi 25 m, eni 12 m. Suvning chuqurligi qancha?

10. Akvariumning bo'yi 40 cm, eni 30 cm, balandligi 35 cm. U to'lishi uchun yana 18 l suv kerak.
    a) Akvariumning bo'sh qismi balandligi necha cm?
    b) Suv sathi necha cm balandlikda?

11. Kuboidning bo'yi 15 cm, eni bo'yidan 5 cm qisqa. Hajmi 1200 cm³. Kuboidning balandligini toping.

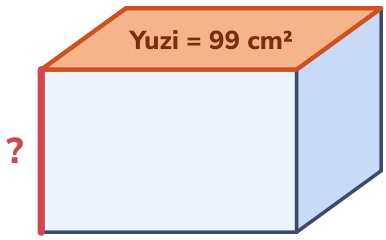
12. Ikki qutining hajmi bir xil. Birinchi quti bo'yi 16 cm, eni 10 cm, balandligi 6 cm. Ikkinchi qutining asosi 12 cm ga 8 cm.
    a) Ikkinchi qutining balandligi qancha?
    b) Qaysi quti balandroq va necha cm ga?

4-QISM

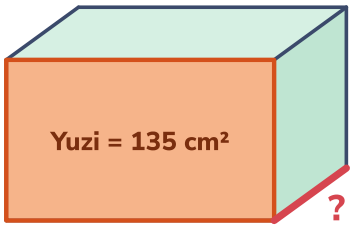
Mustaqil ish

1. Kuboidning noma'lum qirrasini toping.
   a) Hajmi 168 cm³, bo'yi 7 cm, eni 4 cm. b) Hajmi 480 m³, bo'yi 16 m, balandligi 6 m.
   c) Hajmi 96 cm³, eni 4 cm, balandligi 1,5 cm.

2. Rasmga qarab noma'lum qirrani toping.



a) Hajmi 693 cm³



b) Hajmi 1350 cm³

3. Lug'at kitobining hajmi 1800 cm³, bo'yi 20 cm, eni 15 cm. Kitobning qalinligi qancha?

4. Jadvalni to'ldiring.

Kuboid	Bo'yi (cm)	Eni (cm)	Balandligi (cm)	Hajmi (cm³)
K	8	?	5	240
L	?	6	6	396
M	4	5	?	90

5. Hajmi 210 cm³ bo'lgan quti uch xil qo'yildi. Asosining yuzi: 35 cm², 42 cm², 30 cm². Har holatda qutining balandligi qancha?

6. Bo'yi 60 cm, eni 25 cm bo'lgan idishga 45 l suv quyildi. Suv sathi qanday balandlikda?

7. **Xatoni toping.** Kuboidning hajmi 360 cm³, bo'yi 10 cm, eni 4 cm. Sardor balandlikni shunday topdi: $360 : 10 \cdot 4 = 144$ cm. Xato nimada? To'g'ri javob qancha?

8. **Qiyinroq.** Asosi kvadrat bo'lgan ustunning hajmi 540 cm³, kvadrat tomoni 6 cm.
   a) Ustunning balandligi qancha?
   b) Ikkita shunday ustun ustma-ust qo'yildi. Hosil bo'lgan kuboidning balandligi va hajmi qancha?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $5 \cdot 5 \cdot 5 = 125$ m³. b) $9 \cdot 4 \cdot 3 = 108$ cm³. c) $7 \cdot 7 \cdot 7 = 343$ cm³. d) $12 \cdot 5 \cdot 2 = 120$ m³.

2-masala

a) $\frac{120}{5 \cdot 4} = \frac{120}{20} = 6$ cm. b) $\frac{270}{9 \cdot 3} = \frac{270}{27} = 10$ m. c) $\frac{160}{4 \cdot 5} = \frac{160}{20} = 8$ cm.

3-masala

A: $9 \cdot 5 \cdot 2 = 90$. B: $\frac{300}{12 \cdot 5} = \frac{300}{60} = 5$. C: $\frac{540}{15 \cdot 6} = \frac{540}{90} = 6$. D: $8 \cdot 2{,}5 = 20$, $\frac{220}{20} = 11$. E: $\frac{2000}{20 \cdot 20} = \frac{2000}{400} = 5$.

4-masala

a) $40 : (5 \cdot 4) = 40 : 20 = 2$. b) $40 : (10 \cdot 1) = 40 : 10 = 4$. c) $40 : (2 \cdot 2) = 40 : 4 = 10$.

5-masala

$\frac{162}{9 \cdot 6} = \frac{162}{54} = 3$ cm.

6-masala

$\frac{324}{36} = 9$ cm.

7-masala

a) $\frac{560}{80} = 7$ cm (balandligi). b) $\frac{1800}{150} = 12$ m (bo'yi). c) $\frac{630}{105} = 6$ cm (eni).

8-masala

$\frac{360}{60} = 6$ cm, $\frac{360}{90} = 4$ cm, $\frac{360}{24} = 15$ cm. G'ishtchaning qirralari: 15 cm, 6 cm, 4 cm ($15 \cdot 6 \cdot 4 = 360$).

9-masala

Asosi: $25 \cdot 12 = 300$ m². Chuqurligi: $\frac{450}{300} = 1{,}5$ m.

10-masala

a) $18\ \text{l} = 18\,000$ cm³. Asosi: $40 \cdot 30 = 1200$ cm². Bo'sh qism: $18\,000 : 1200 = 15$ cm. b) $35 - 15 = 20$ cm.

11-masala

Eni: $15 - 5 = 10$ cm. Balandligi: $\frac{1200}{15 \cdot 10} = \frac{1200}{150} = 8$ cm.

12-masala

a) Hajmi: $16 \cdot 10 \cdot 6 = 960$ cm³. Balandligi: $\frac{960}{12 \cdot 8} = \frac{960}{96} = 10$ cm. b) Ikkinchi quti $10 - 6 = 4$ cm ga balandroq.

Mustaqil ish

1-masala

a) $\frac{168}{7 \cdot 4} = \frac{168}{28} = 6$ cm (balandligi). b) $\frac{480}{16 \cdot 6} = \frac{480}{96} = 5$ m (eni). c) $4 \cdot 1{,}5 = 6$, $\frac{96}{6} = 16$ cm (bo'yi).

2-masala

a) $\frac{693}{99} = 7$ cm. b) $\frac{1350}{135} = 10$ cm.

3-masala

$\frac{1800}{20 \cdot 15} = \frac{1800}{300} = 6$ cm.

4-masala

K: $\frac{240}{8 \cdot 5} = \frac{240}{40} = 6$. L: $\frac{396}{6 \cdot 6} = \frac{396}{36} = 11$. M: $\frac{90}{4 \cdot 5} = \frac{90}{20} = 4{,}5$.

5-masala

$\frac{210}{35} = 6$ cm, $\frac{210}{42} = 5$ cm, $\frac{210}{30} = 7$ cm.

6-masala

$45\ \text{l} = 45\,000$ cm³. Asosi: $60 \cdot 25 = 1500$ cm². Suv sathi: $45\,000 : 1500 = 30$ cm.

7-masala

Sardor 360 ni 10 ga bo'lib, keyin 4 ga ko'paytirgan. Hajmni ikki qirraning ko'paytmasiga bo'lish kerak: $\frac{360}{10 \cdot 4} = \frac{360}{40} = 9$ cm.

8-masala

a) Asosi: $6 \cdot 6 = 36$ cm². Balandligi: $\frac{540}{36} = 15$ cm. b) Balandligi $15 + 15 = 30$ cm, hajmi $36 \cdot 30 = 1080$ cm³.